

An LLM Agent for the Selection, Validation and Interpretation of Statistical Methods

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Programme	Capstone Project
Document	Presentation Script
Date	21 September 2026

Twenty-two slides, about 21 minutes spoken. Words in grey brackets are cues, not lines to say.

Slide 1 · Title

about 0.5 minutes

Hello everyone. My name is Vishal Dhinesh Kumar, and my capstone project is called AI Statistician.

The question I set out to answer is this. If you give an AI a proper step-by-step procedure for doing statistics, does it choose the right method more often than if you just give it the tools and let it work things out on its own?

That's what I'm going to walk you through today.

Slide 2 · Problem statement

about 2 minutes

So let me start with the problem, because everything else in this project comes back to it.

If you ask an AI to analyse a dataset, it will always give you an answer. And that answer usually looks good. It's well written, it uses the right terms, and the arithmetic is correct.

But none of that tells you whether the method was actually the right one to use.

Whether a statistical method is valid depends on how the data was collected. Are the observations independent of each other? Is the same person measured twice? Are the rows in time order? These are facts about the study design. And the important thing is, you can't see them in the data file. A spreadsheet only shows you numbers.

[point to the three boxes] This creates three problems.

The first one is the most dangerous. The model can pick a method that assumes something the data doesn't satisfy. Every number will still add up, so it looks completely fine. But the conclusion is wrong.

The second is that the model can write a number that no calculation ever produced. It reads like a real result, and the only way to catch it is to redo the whole analysis yourself.

And the third is that nothing forces the model to stop and say, this data doesn't fit any method I have. Giving it tools lets it calculate things. It doesn't make it check whether it should.

That's the gap I wanted to work on.

Slide 3 · A motivating case

about 1.5 minutes

To make this concrete, here's an example from my benchmark.

This is the Seoul bike sharing dataset from the UCI repository. It has 8,760 rows, one for every hour of a year. The question is whether temperature is related to how many bikes get rented.

If you look at this table, it looks like a textbook case. You'd probably reach for a correlation, or a Poisson regression because it's count data. And that's exactly what the systems did.

The problem is that both of those methods assume every observation is independent. Here they're not. Two o'clock looks a lot like one o'clock, because it's the same day, the same weather, the same commute. Each hour is carrying information about the hours next to it.

And there's nothing in the file that tells you this. You only know it because you know how the data was collected.

So in my system, that information comes in through what I call a design card, which describes how the study was set up. The system is scored on whether it actually uses it.

For this case, the correct answer is not to model it at all. It should decline.

Slide 4 • Research question and hypotheses

about 1 minute

So this is my research question. Does an explicit decision protocol help an AI choose valid methods, compared with just asking it directly, and compared with giving it tools but no protocol?

I tested three hypotheses. H1 is about selection, whether the protocol picks the right method more often. H2 is about abstention, whether it declines more reliably when no method fits. And H3 is about provenance, whether it stops made-up numbers from getting into the final report.

One thing I want to point out. For H1, the main comparison is C against B, not C against A. That's on purpose. B has exactly the same tools and the same report format as C. The only thing different is the procedure. So if C beats B, I know it's the procedure that made the difference, not the tools.

Slide 5 · Scope and design decisions

about 1 minute

Before the results, a few design decisions that make this testable.

[point to the left] This is the method library. I limited it to fourteen methods, grouped by the kind of question. Two groups, three or more groups, categorical data, association, and regression. And there's a fifteenth option, which is to abstain.

I kept it closed on purpose. If the model could invent any method it wanted, there'd be no clear way to mark whether its choice was right. It would turn into grading essays.

I also didn't give it a general tool to run any code it likes. If I had, it could skip the procedure completely, and then the three systems wouldn't be doing comparable work.

And the designs I didn't cover, like repeated measures or time series, I didn't just leave out. They're in the benchmark as cases where the right answer is to decline.

Slide 6 · Architecture

about 1 minute

This is how the three systems are built.

A is the direct version, with no tools. B has the tools but no procedure. And C is the structured agent, which has to check the design first.

[point to the bottom box] The important part is down here. All three share the same core. The same tool registry, result store, report format and trace log. The only thing that changes between them is the control flow, meaning the order they do things in.

I built it this way so the comparison is fair by design. There's no part of the code where A or B gets a weaker version of something. So if the results are different, it has to be because of the control flow.

Slide 7 · The decision protocol

about 1.25 minutes

This is what that control flow looks like for system C. There are seven steps.

It reads the question, checks the design, lists the methods that design allows, runs only the checks that could actually change the choice, then either picks a method or declines, runs it, and verifies the report at the end.

[point to steps 2 and 5] The two in green are the ones that matter most.

Design validation happens before anything else. So if the design has a problem, say the rows aren't independent, those methods are removed before any diagnostic even runs. That means a normality test that happens to look fine can never push the system into a method the design has already ruled out.

And step five is where it's allowed to say no.

I want to be clear about one thing. System B can run every one of these checks. Nothing is stopping it. What B doesn't have is the rule that says check the design first, and the permission to stop.

Slide 8 · The provenance contract

about 1 minute

The other big design decision is about numbers. The idea is that the model isn't allowed to write a number itself.

Every time a real calculation runs, the result is stored with a label. The report format rejects any number typed straight into the text. So instead, the model writes a reference to the label. At the end, every reference is replaced with the real stored value. And if the model refers to something that was never calculated, the whole run fails.

So instead of checking a finished report for made-up numbers, it just isn't possible to put one in.

I do want to be honest about what this means. For system C, numerical accuracy is guaranteed by how it's built. It's not something I measured and discovered. So I'm not going to present it as a result. What I can measure is how often the model tried to reference something that didn't exist.

Slide 9 · The benchmark

about 1 minute

To test all this, I built a benchmark of 64 cases. 52 of them have a correct method, and 12 are cases where the right answer is to decline. 48 of the 64 are held out, which I'll explain on the next slide.

None of these were written by hand. Each case is generated from a definition. For the synthetic ones, the correct answer is based on the data that actually came out, not what I meant to generate. So if I asked for normal data and it came out skewed, the label follows the skewed data.

[point to the right] Then there are four public datasets from UCI.

The reason this matters is that most of the answer key isn't my opinion. 48 of the labels come from how the data was built. The 16 public ones I checked against the real values, and that check found one label that was wrong, which I fixed.

Slide 10 · Evaluation protocol

about 1 minute

This slide is about how I kept the evaluation fair.

The biggest risk in a project like this is that you end up tuning your system to the test without realising it. So the 48 held-out cases were locked away. I didn't look at them while I was building. I also fixed the instructions given to the model and recorded a fingerprint of them before the held-out run, so they couldn't change afterwards.

Each case was run twice for each system. That's 288 runs in total, all on the same model.

On the right are the metrics. Accuracy with confidence intervals, how often it declined correctly, whether the report passed the provenance check, and where each system failed.

For the statistics, I compared the systems case by case using McNemar's test, because every case was answered by all three systems.

Slide 11 · Result: accuracy

about 1 minute

So now the results.

This is how often each system chose a valid method. A, the direct version, got 75 percent. B, with tools only, got 67 percent. And C, with the procedure, got 89 percent.

C beat B by 25 percentage points. My target at the start was 10 points, so that's well above it.

[point to the B bar] But the part I found most interesting is B. B has tools, and it actually did worse than A, which has no tools at all.

So giving the model tools on their own didn't help. It was the procedure that made the difference.

Slide 12 · Result: which differences are real

about 1.25 minutes

Now, which of these differences are actually real? That's what this slide shows.

A solid green line with a filled dot means the difference is statistically significant. A dashed grey line with a hollow dot means it isn't.

C over B, the main comparison, has a p-value of 0.0018. So that one is established.

C over A is a 12.5 point difference, but the p-value is 0.07. That's not below 0.05, so I can't claim that C is better than A. It's pointing in C's favour, but with 48 cases I don't have enough data to rule out chance. So I'm reporting it as unresolved.

B over A isn't significant either.

[slow down here] I gave this its own slide because I think it's important to be clear about where the evidence stops. If I only believed my results when they agreed with me, they wouldn't really be results.

Slide 13 · Result: recognising an invalid design

about 0.75 minutes

This is the safety result, and it's the clearest difference in the whole study.

These are the cases where every available method is invalid, so the right answer is to decline.

C declined correctly 78 percent of the time. B declined zero times. Every single time, B went ahead and fitted a method to data that didn't support it.

That's exactly the failure this project was trying to prevent. So H2 is supported.

Slide 14 · Result: could the report be grounded

about 0.75 minutes

This one is about whether each system could report its results without making anything up.

System A had 43 out of 96 reports rejected. That means it either wrote a number that no calculation produced, or it mentioned a check it never actually ran.

You might notice B is at 100 percent here. That's because B has tools and uses them, so its numbers are real. B's problem isn't making up numbers. It's picking the wrong method in the first place.

The main point is that none of these made-up numbers reached a reader. They were all caught. So H3 is supported.

Slide 15 · Result: where each system fails

about 0.75 minutes

This chart shows where each system actually goes wrong.

The solid part of each bar is design validation failures. That means using a method on data where the design doesn't allow it.

B has 20 of those. C has 4.

So the procedure cut down exactly the mistake it was built to stop. Most of what's left for C is in diagnostic reasoning, which is a different and smaller problem.

Slide 16 · Result: coverage and accuracy

about 0.75 minutes

A fair question here is whether C just refuses to answer more often, and that's why it looks better.

This chart answers that. Along the bottom is how often each system chose to answer. Up the side is how accurate it was on the ones it did answer.

C answered 85 percent of cases and was right on 87 percent of those. A and B answered almost everything, 95 and 98 percent, but they were right less often.

So C gave up some coverage, but it got accuracy back for it. That's what you'd expect if it's declining the right cases, and not just declining at random.

Slide 17 · Hypotheses revisited

about 0.75 minutes

So going back to my hypotheses.

H1, selection, is supported. C beat B by 25 points.

H2, abstention, is supported. C declined correctly 78 percent of the time, and B never did.

H3, provenance, is supported. 51 reports were rejected, and none of them reached a reader.

The comparison between C and A is still unresolved. I left that on the slide on purpose, because taking it off would make the findings look stronger than they are.

Slide 18 · Two defects in my own measurement

about 1.25 minutes

This part isn't about the system. It's about how I measured it, and two mistakes I found.

The first one. 51 runs ended with an error, and the obvious thing to do is throw them out. But when I looked at them, none were technical failures. Every one had already chosen a method before its report was rejected, and 37 of them had chosen correctly. If I'd thrown them out, system A would have looked 6 points better than it really is. So I kept them.

The second one. Each case was run twice, and sometimes the two runs disagreed. I needed a way to break those ties, and the way I first did it changed from one session to the next. So the exact same data gave me different p-values, anywhere from 0.016 to 0.125. That's on both sides of 0.05. I fixed it so ties always break the same way, and added a test so it stays fixed.

Neither of these showed up as an error. I only found them by going through the results carefully before building any tables.

Slide 19 · Limitations

about 1 minute

These are the limitations.

The held-out results are for one model, Claude Haiku 4.5. I'm not claiming anything about larger models without testing them.

The sample is 48 cases with two runs each. That's enough to show the main result, but not enough to settle C against A.

I also tried to score how good each system's written interpretation was, with two people scoring blind. That didn't work. The two scorers didn't agree with each other. The agreement score was minus 0.044, which is actually worse than chance. When I looked into why, the scoring rubric could be read in two different ways. So I rebuilt it, and I've left that metric out rather than report a number I can't trust.

There's also no second independent reviewer for the public data labels, and a few study designs are outside what the library covers.

Slide 20 · Contributions

about 0.75 minutes

So what does this project add?

First, controlled evidence that it's the procedure, and not just giving the AI tools, that makes it choose valid methods.

Second, a benchmark that actually rewards declining, so knowing when to stop becomes something you can measure.

And third, a way of reporting results where a number with no real calculation behind it can't exist.

If I had to put it in one line: structure, not tools, is what makes an AI a reliable statistician.

[pause]

Slide 21 · Future work

about 0.5 minutes

For future work, the first thing I'd do is run a bigger held-out set, so I can actually settle the question of C against A.

I'd also like to test it on larger models, to see whether the procedure still helps as the model gets more capable. And I'd run a second round of the interpretation scoring with the new rubric, and extend the library to cover designs like repeated measures and time series.

Slide 22 · Thank you

about 0.5 minutes

Everything I've shown today is online. The first link is a live explorer where you can see every case, every answer each system gave, and every step it took. The second is the full code and data on GitHub.

If there's time, I'm happy to open the bike sharing example and show C declining while the other two go ahead anyway.

Thank you. I'm happy to take any questions.

If I get asked

the questions most likely to come up

If numbers can't be made up in system C, isn't its accuracy on numbers automatic?

Yes, it is, and that's why I don't count it as a result. It's guaranteed by how the system is built. What I actually measure is how often the model tried to reference something that was never calculated.

Why is your main comparison C against B and not C against A?

Because B has exactly the same tools and report format as C. The only difference is the procedure. If I compared C with A, I'd be mixing up the effect of the procedure with the effect of having tools.

C over A isn't significant. So is the procedure really better than nothing?

I can't say that yet. The difference points in C's favour, but with 48 cases it's not enough to rule out chance, so I've reported it as unresolved. The main comparison, against B, is significant at 0.0018.

How do you know your answer key is right?

48 of the 64 labels come from how the data was generated, so they're not my judgement. For the 16 public datasets, I checked each label against the real data, and that found one mistake, which I corrected.

Why only two runs per case?

It came down to resources. Two runs were enough to show the main result and to measure how consistent the systems were, but not enough to fully describe the variation. More runs is the first thing I'd add.

Why did you keep the runs that ended in errors?

Because none of them were technical failures. Each one had already picked a method before its report was rejected, and 37 had picked correctly. Throwing them out would have made system A look 6 points better than it actually is.

Couldn't system B just do what C does?

It could, in theory. It has every tool C has. What it doesn't have is the rule to check the design first, and permission to stop. And the results show that without those, it doesn't do it on its own.

What happened with the interpretation scoring?

The two scorers didn't agree, the agreement was minus 0.044, which is worse than chance. The rubric could be read two ways. So I rebuilt it and left the metric out, because a number from a scoring method that doesn't work would be misleading.